

Towards Efficient and Interpretable Medical Concept Representation via Ontology-driven Residual Vector Quantization

Hang Lv
Fuzhou University
Fuzhou, China
lvhangkenn@gmail.com

Kaisong Zhang
Fuzhou University
Fuzhou, China
832403326@fzu.edu.cn

Yanchao Tan[†]
Fuzhou University
Fuzhou, China
yctan@fzu.edu.cn

Xing Chen[†]
Fuzhou University
Fuzhou, China
chenxing@fzu.edu.cn

Abstract

Medical concepts, the core entities in Electronic Health Records (EHRs), provide essential inputs for clinical decision-making systems. However, most existing healthcare models still rely on massive concept-specific embedding tables, resulting in substantial memory overhead. Recent studies compress medical concepts into discrete code sequences for memory efficiency, but their flat semantic quantization fails to explicitly encode the hierarchical structure of medical ontologies, thereby limiting clinical interpretability. To this end, we propose MedRQ, an ontology-driven residual vector quantization framework that aligns discrete codes with multi-level clinical ontologies. By incorporating hierarchical supervision into the quantization process, MedRQ generates compact and ontology-consistent concept representations that generalize seamlessly across healthcare prediction tasks. Experiments on two real-world EHR datasets demonstrate that MedRQ significantly outperforms state-of-the-art baselines while reducing memory usage.

CCS Concepts

• Applied computing → Health informatics.

Keywords

Representation Learning, Medical Concept Ontology, Residual Vector Quantization

ACM Reference Format:

Hang Lv, Kaisong Zhang, Yanchao Tan[†], and Xing Chen[†]. 2026. Towards Efficient and Interpretable Medical Concept Representation via Ontology-driven Residual Vector Quantization. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3774904.3792930>

1 Introduction

Electronic Health Records (EHRs), as a central data foundation for modern healthcare prediction, provide rich information for patient condition assessment and clinical decision-making [9, 11, 12, 15]. The cornerstone of EHRs is medical concepts (e.g., diagnoses, procedures, and medication), which originate from standardized

clinical classification systems (e.g., ICD-9¹ and ATC²) that structure medical knowledge into multi-level ontologies.

Although learning effective representations of these medical concepts is essential for reliable prediction, the massive number of concepts incorporated into clinical models poses a major challenge. For example, the ICD-9 system alone contains over 14,000 diagnosis codes and 3,900 procedure codes. Treating each concept as an independent embedding results in substantial memory overhead, hindering their practicality in real-world clinical settings.

Recent studies try to compress medical concepts into discrete code sequences, replacing the large embedding tables required by traditional models [1, 11] and improving computational efficiency. For example, UDC [14] maps diagnosis embeddings to a set of discrete codes through a Residual-Quantized Variational AutoEncoder (RQ-VAE) [4]. Despite their effectiveness in reducing memory cost, these methods typically learn flat codes driven solely by semantic similarity. Such unsupervised quantization cannot guarantee that the learned codes reflect the multi-level organization of medical ontologies. For instance, ICD-9 codes 401 *Essential Hypertension* and 405 *Secondary Hypertension* share textual similarities, but they differ in etiology and treatment pathways. Without explicit ontology guidance, the learned discrete medical concept representations may lose discriminability and lead to inaccurate clinical predictions [7].

In this paper, we propose MedRQ, an efficient and interpretable medical concept representation learning framework, which directly integrates hierarchical ontology supervision into quantized concept codes for enhanced healthcare prediction. Specifically, Stage 1 designs an Ontology-driven RQ-VAE to align discrete code sequences with multi-level ontology structures, yielding compact and ontology-consistent representations. Stage 2 leverages these concept embeddings for healthcare prediction, consistently outperforming state-of-the-art baselines. Notably, the learned quantized codes explicitly encode ontology structure, making them task-agnostic and potentially applicable to a variety of clinical prediction tasks.

2 Methodology

2.1 Notation and Problem Formulation

Medical Concept Ontology. A medical concept ontology provides a hierarchical organization of clinically related concepts, ranging from general to specific entities. Each concept type (e.g., diagnosis) is associated with its own ontology (e.g., ICD-9). We denote the ontology as $\mathcal{G} = \{\mathcal{G}_l\}_{l=1}^L \cup C$, where $L + 1$ is the total depth of the hierarchy, $g_l^c \in \mathcal{G}_l$ is the l -th level (higher-level) concept, and $c \in C = \mathcal{G}_{L+1}$ is the leaf-level concept typically recorded in EHRs.

[†] Yanchao Tan and Xing Chen are the corresponding authors.



This work is licensed under a Creative Commons Attribution 4.0 International License. *WWW '26, Dubai, United Arab Emirates*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792930>

¹<https://www.cdc.gov/nchs/icd/icd9cm.htm>

²<https://www.who.int/tools/atc-ddd-toolkit/atc-classification>

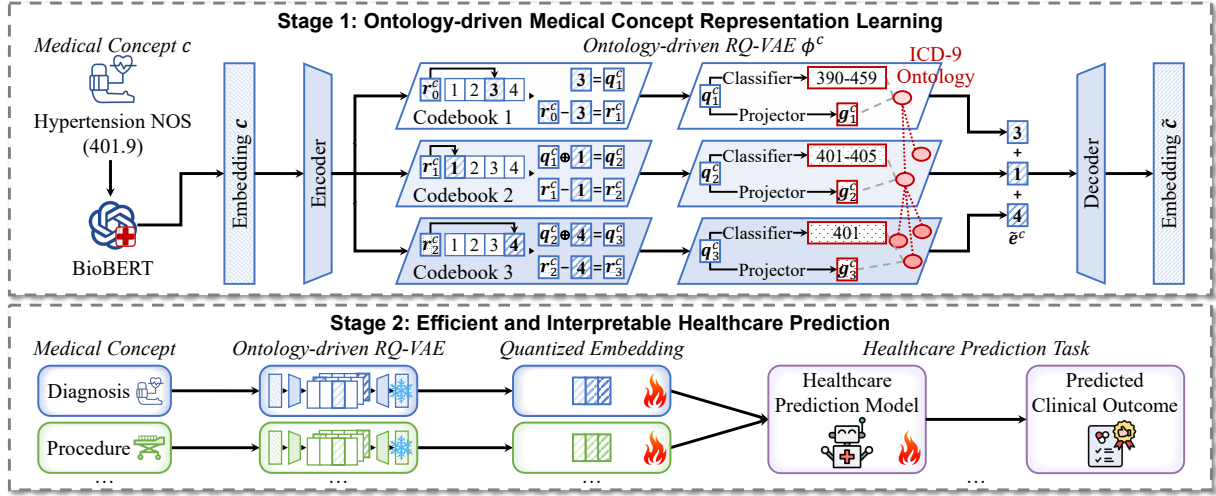


Figure 1: The MedRQ overall framework. MedRQ first learns medical concept representations via an Ontology-driven RQ-VAE (Stage 1), and then performs efficient and interpretable healthcare prediction using the quantized embeddings (Stage 2).

Problem Formulation. Our goal is to train a medical concept representation learning model $\phi^c(\cdot)$ for healthcare prediction tasks (e.g., medication recommendation). Given a patient's visit sequence $\mathcal{V} = \{v_t\}_{t=1}^T$, where each visit v_t contains a set of various medical concepts c , the prediction task aims to infer clinical outcomes. In this paper, we focus on medication recommendation for the current visit v_T . Note that the learned representations remain potentially applicable to other prediction tasks (e.g., diagnosis prediction).

2.2 Stage 1: Ontology-driven Medical Concept Representation Learning

To learn efficient and clinically interpretable representations of medical concepts, Stage 1 introduces an Ontology-driven RQ-VAE that explicitly integrates a base RQ-VAE with hierarchical ontology supervision (shown in Figure 1).

Base RQ-VAE. For each medical concept c with a textual description (e.g., *Hypertension NOS*), we first generate its semantic embedding $\tilde{c}$ via a pre-trained BioBERT [5]. The semantic embedding is then mapped to a latent vector $x \in \mathbb{R}^{1 \times \dim}$ through an encoder. We subsequently employ an RQ-VAE [4] to quantize x into a compact discrete code sequence via L -level codebooks, where L denotes the sequence length. Specifically, the quantizer selects the most similar code embedding e_l^c from the l -th level codebook $\mathcal{E}_l = \{e_{l,k}^c\}_{k=1}^N$ based on the current semantic residual and updates the residual:

$$e_l^c = \arg \min_k \|r_{l-1}^c - e_{l,k}^c\|, \quad r_l^c = r_{l-1}^c - e_l^c, \quad (1)$$

where e_l^c is the selected code index at level l , r_{l-1}^c is the residual from the previous level, and the initial residual is set as $r_0^c = x$. After the recursive quantization, we obtain the discrete code sequence $(e_1^c, \dots, e_L^c)$ for concept c and its corresponding quantized embedding $\tilde{e}^c = \sum_{l=1}^L e_l^c$. The quantized embedding $\tilde{e}^c$ is then decoded to yield the reconstructed semantic embedding $\tilde{c}$. Moreover, the RQ-VAE is trained with the standard objective $\mathcal{L}_{\text{RQ-VAE}} = \|c - \tilde{c}\|^2 + \sum_{l=1}^L \|\text{sg}[r_{l-1}^c] - e_l^c\|^2 + \lambda \|r_{l-1}^c - \text{sg}[e_l^c]\|^2$, where $\text{sg}[\cdot]$ denotes the stop-gradient operator and λ controls the balance between optimizing the code embeddings and the encoder.

Ontology-driven RQ-VAE. To explicitly integrate structured ontology information into the quantization process, we design a dual-supervision mechanism that aligns the discrete codes of each medical concept c with its higher-level ontology concepts $\{g_l^c\}_{l=1}^L$ across the L -level codebooks of RQ-VAE. This mechanism consists of two complementary losses, i.e., alignment loss and discrimination loss, which are defined as follows:

$$\mathcal{L}_{\text{Align}} = \sum_{l=1}^L -\log \frac{\exp(\text{sim}(q_l^c, \mathcal{P}_l(g_l^c))/\tau)}{\sum_{k \in \mathcal{G}_l} \exp(\text{sim}(q_l^c, \mathcal{P}_l(g_l^k))/\tau)}, \quad (2)$$

where the embedding $q_l^c = [e_1^c \| \dots \| e_l^c]$, g_l^c is the semantic embedding of the l -th level concepts from BioBERT, $\text{sim}(\cdot, \cdot)$ is cosine similarity, $\mathcal{P}_l(\cdot)$ is the l -th level projector, and τ is a temperature hyperparameter. This loss enforces semantic consistency between the learned codes and the ontology hierarchy. Furthermore, since higher-level concepts may share very similar clinical semantics, relying solely on semantic alignment can reduce the discriminability among the corresponding code representations. To preserve clear separation, we classify each embedding q_l^c into its correct ontology category g_l^c via the following discrimination loss:

$$\mathcal{L}_{\text{Disc}} = \sum_{l=1}^L \text{CrossEntropy}(\mathcal{F}_l(q_l^c), g_l^c), \quad (3)$$

where $\mathcal{F}_l$ denotes the l -th level MLP classifier. Finally, the overall training objective of Ontology-driven RQ-VAE $\phi^c(\cdot)$ is as follows:

$$\mathcal{L}_{\text{Stage 1}} = \mathcal{L}_{\text{RQ-VAE}} + \gamma \mathcal{L}_{\text{Onto}} = \mathcal{L}_{\text{RQ-VAE}} + \gamma (\mathcal{L}_{\text{Align}} + \mathcal{L}_{\text{Disc}}), \quad (4)$$

where γ controls the strength of ontology-driven supervision.

2.3 Stage 2: Efficient and Interpretable Healthcare Prediction

With the compact and ontology-consistent codes learned in Stage 1, Stage 2 directly leverages these discrete representations to enable efficient and accurate healthcare prediction.

Quantized Medical Concept Embedding. For each medical concept c , we retrieve its discrete code sequence using the trained Ontology-driven RQ-VAE $\phi^c(\cdot)$ and construct its learnable quantized embedding as $q_L^c = [e_1^c \| \dots \| e_L^c]$. Based on the above unified

Table 1: Statistics of the datasets used in our experiments.

Dataset	MIMIC-III	MIMIC-IV
# patients / # visits	6350 / 15031	61264 / 163877
# diag. / # proc. / # med.	1903 / 1409 / 131	2000 / 11056 / 131
Avg. / Max # visits	2.3671 / 29	2.6749 / 70
Avg. / Max # diag. per visit	10.2266 / 127	8.2343 / 270
Avg. / Max # proc. per visit	3.8244 / 50	2.3579 / 95
Avg. / Max # med. per visit	11.4361 / 65	6.5055 / 72

formulation, we obtain the embeddings q_L^d , q_L^p , and q_L^m for diagnoses, procedures, and medications, respectively.

Medication Recommendation. Given a patient’s current visit v_T , we generate the visit representations by aggregating the obtained embeddings of diagnoses, procedures, and historical medications $v_T = \{v_T^d, v_T^p, v_T^m\}$. To further capture temporal dependencies across visits, we employ three GRUs to encode the longitudinal visit sequences, i.e., $h_T^d = \text{GRU}^d(\{v_t^d\}_{t=1}^T)$, $h_T^p = \text{GRU}^p(\{v_t^p\}_{t=1}^T)$, and $h_{T-1}^m = \text{GRU}^m(\{v_t^m\}_{t=1}^{T-1})$. The resulting latent embeddings are concatenated to form the patient representation $z_T = [h_T^d \| h_T^p \| h_{T-1}^m]$. The medication prediction for the current visit v_T is then obtained as $\hat{y}_T = \mathcal{F}_{\text{Rec}}(z_T)$, where $\mathcal{F}_{\text{Rec}}$ is a multi-layer perceptron. Consistent with the prior studies [11, 12, 15], we train MedRQ for medication recommendation using three different loss functions as follows:

$$\begin{aligned}
 \mathcal{L}_{\text{BCE}} &= - \sum_{i=1}^{|\mathcal{M}|} [\mathbf{y}_{T,i} \log \hat{\mathbf{y}}_{T,i} + (1 - \mathbf{y}_{T,i}) \log(1 - \hat{\mathbf{y}}_{T,i})], \\
 \mathcal{L}_{\text{Multi}} &= \sum_{\{i | \mathbf{y}_{T,i}=1\}} \sum_{\{j | \mathbf{y}_{T,j}=0\}} \frac{\max\{1 - (\hat{\mathbf{y}}_{T,i} - \hat{\mathbf{y}}_{T,j}), 0\}}{|\mathcal{M}|}, \\
 \mathcal{L}_{\text{DDI}} &= \sum_{i=1}^{|\mathcal{M}|} \sum_{j=1}^{|\mathcal{M}|} \mathbf{A}_{ij} \cdot \hat{\mathbf{y}}_{T,i} (\hat{\mathbf{y}}_{T,j})^T,
 \end{aligned} \quad (5)$$

where $\mathbf{y}_T \in \{0, 1\}^{|\mathcal{M}|}$ is the ground-truth medications for the patient’s current visit, $|\mathcal{M}|$ is the total number of medications, and $\mathbf{A}$ is a symmetric binary Drug-Drug Interaction (DDI) adjacency matrix. The final training objective of the recommendation prediction can be formulated as $\mathcal{L}_{\text{Stage 2}} = \alpha \mathcal{L}_{\text{BCE}} + (1 - \alpha) \mathcal{L}_{\text{Multi}} + \beta \mathcal{L}_{\text{DDI}}$, where α and β balance the contributions of the different losses.

3 Experiments

3.1 Experimental Settings

Datasets and Evaluation Protocols. We evaluate MedRQ on two real-world EHR datasets, MIMIC-III [2] and MIMIC-IV [3], which contain patients with at least two visits and provide standardized medical concepts (ICD-9 diagnoses/procedures, and ATC medications). The statistics are summarized in Table 1. We use Jaccard Similarity (Jaccard), Average F1 (F1), Precision Recall AUC (PRAUC), Drug-Drug Interaction Rate (DDI), and Average Number of Medications (# Med.) as metrics, consistent with [9, 12, 15].

Methods for Comparison. We compare against 10 representative state-of-the-art baselines from two main categories: (1) Instance-based methods (LR and ECC [8]); (2) Longitudinal methods (RETAIn [1], LEAP [13], MICRON [10], GAMENet [9], SafeDrug [11], MoleRec [12], DEPOT [15], and UDC [14]).

Implementation Details. We split training, validation, and test sets by 2/3, 1/6, and 1/6, consistent with [11, 12, 15]. For Stage 1,

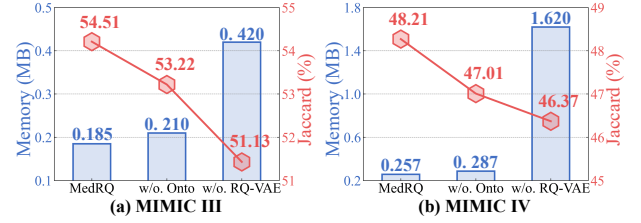


Figure 2: Efficiency results of MedRQ and its ablation variants.

we set the embedding dimension dim to 64 and employ a 3-layer MLP with GELU activations as the encoder and decoder. The RQ-VAE uses $L = 3$ codebook levels, each with $N = 256$ size, and all codebooks are initialized using K-Means clustering. The key loss hyperparameters are set to $\lambda = 0.25$, $\gamma = 1.0$, and $\tau = 0.07$. For Stage 2, the frozen Ontology-driven RQ-VAE provides pre-computed code sequences as inputs. We use a single-layer GRU with a hidden dimension of 192. The loss weight α and β are set to 0.95 and 1.0. All experiments are performed by one NVIDIA RTX 4090 GPU³.

3.2 Overall Performance Comparison

As shown in Table 2, MedRQ achieves the best overall performance on MIMIC-III and MIMIC-IV across all accuracy metrics (i.e., Jaccard, F1, and PRAUC). This demonstrates that the efficient and interpretable medical concept representations learned through our Ontology-driven RQ-VAE can effectively enhance healthcare prediction tasks. Compared with the second-best performance (e.g., DEPOT), the performance gains of MedRQ achieve up to 2.25% with Jaccard on MIMIC-III. Notably, although UDC benefits from discrete code compression, its flat codes ignore ontology hierarchy and thus lack clinical discriminability, yielding lower performance than MedRQ. For DDI safety, SafeDrug leads with its DDI-controllable loss function. In comparison, MedRQ significantly improves all accuracy metrics (achieved up to 12.81% in Jaccard on MIMIC-III) while increasing the DDI rate by only 0.001 on average, reflecting a practical balance between prediction accuracy and safety.

3.3 Model Efficiency Analysis

Figure 2 compares MedRQ with two ablation variants: w/o. Onto, which removes the ontology-driven loss $\mathcal{L}_{\text{Onto}}$, and w/o. RQ-VAE, which replaces our quantized codes with standard randomly initialized embedding tables. Across both datasets, MedRQ achieves the lowest memory usage while maintaining the highest Jaccard accuracy. Compared with the w/o. RQ-VAE variant, MedRQ reduces storage overhead by up to 5.3× on MIMIC-IV, where the larger number of medical concepts increases the efficiency gains of our compact quantized representations. This highlights the scalability of MedRQ in large-scale clinical datasets. Moreover, the accuracy improvement of MedRQ over w/o. Onto show that explicitly integrating ontology hierarchy yields more clinically coherent and discriminative representations, consistent with the findings in Section 3.2.

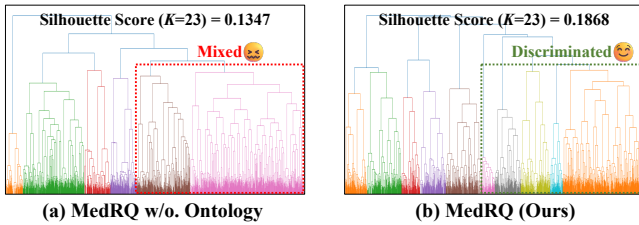
3.4 Interpretable Case Studies

To highlight the advantages of incorporating structured ontology information into the quantization process, we visualize the learned

³The full code for this work is available: <https://github.com/lvhangkenn/MedRQ>.

Table 2: Experimental results (%) on two EHR datasets. The best performances are highlighted in boldface and the second-best results are underlined. Ground-truth # of Med. on the test sets is 19.79 for MIMIC-III and 11.98 for MIMIC-IV, respectively.

Dataset	MIMIC-III					MIMIC-IV				
Method	Jaccard $\uparrow$	F1 $\uparrow$	PRAUC $\uparrow$	DDI $\downarrow$	# Med.	Jaccard $\uparrow$	F1 $\uparrow$	PRAUC $\uparrow$	DDI $\downarrow$	# Med.
LR	46.47 $\pm$ 0.12	62.65 $\pm$ 0.18	74.72 $\pm$ 0.13	8.09 $\pm$ 0.09	15.85	41.01 $\pm$ 0.13	55.89 $\pm$ 0.15	67.17 $\pm$ 0.19	7.76 $\pm$ 0.07	10.03
ECC	45.56 $\pm$ 0.16	61.67 $\pm$ 0.14	71.92 $\pm$ 0.19	8.18 $\pm$ 0.12	15.48	39.27 $\pm$ 0.18	53.74 $\pm$ 0.13	66.91 $\pm$ 0.18	7.83 $\pm$ 0.07	7.89
RETAIN	45.37 $\pm$ 0.13	61.74 $\pm$ 0.16	72.19 $\pm$ 0.12	8.46 $\pm$ 0.07	19.96	41.64 $\pm$ 0.12	56.87 $\pm$ 0.20	67.12 $\pm$ 0.13	8.09 $\pm$ 0.09	10.53
LEAP	43.28 $\pm$ 0.13	59.86 $\pm$ 0.18	64.65 $\pm$ 0.13	7.61 $\pm$ 0.06	17.82	38.98 $\pm$ 0.17	54.05 $\pm$ 0.13	54.36 $\pm$ 0.18	7.28 $\pm$ 0.07	9.78
MICRON	48.19 $\pm$ 0.12	64.03 $\pm$ 0.18	72.97 $\pm$ 0.14	7.54 $\pm$ 0.10	18.71	42.77 $\pm$ 0.13	57.68 $\pm$ 0.19	67.33 $\pm$ 0.13	7.19 $\pm$ 0.09	11.70
GAMENet	47.83 $\pm$ 0.15	63.79 $\pm$ 0.17	72.48 $\pm$ 0.13	8.52 $\pm$ 0.06	24.25	42.19 $\pm$ 0.19	57.45 $\pm$ 0.14	66.29 $\pm$ 0.18	8.17 $\pm$ 0.08	15.54
SafeDrug	48.32 $\pm$ 0.18	64.34 $\pm$ 0.12	72.61 $\pm$ 0.19	6.91$\pm$0.04	18.01	43.57 $\pm$ 0.19	58.73 $\pm$ 0.14	65.43 $\pm$ 0.19	6.63$\pm$0.04	10.85
MoleRec	53.03 $\pm$ 0.13	68.47 $\pm$ 0.20	77.75 $\pm$ 0.14	7.34 $\pm$ 0.04	20.93	46.74 $\pm$ 0.13	61.94 $\pm$ 0.17	70.67 $\pm$ 0.19	6.94 $\pm$ 0.02	11.85
DEPOT	<u>53.31$\pm$0.15</u>	<u>68.68$\pm$0.13</u>	78.20 $\pm$ 0.18	7.22 $\pm$ 0.04	20.37	47.09 $\pm$ 0.12	62.23 $\pm$ 0.19	71.11 $\pm$ 0.11	7.02 $\pm$ 0.09	12.07
UDC	52.21 $\pm$ 0.21	67.74 $\pm$ 0.14	<u>78.24$\pm$0.11</u>	7.29 $\pm$ 0.05	20.05	<u>47.81$\pm$0.16</u>	<u>63.11$\pm$0.12</u>	<u>71.45$\pm$0.15</u>	7.12 $\pm$ 0.07	12.15
MedRQ	54.51$\pm$0.12	69.73$\pm$0.16	78.86$\pm$0.18	7.00 $\pm$ 0.07	21.16	48.21$\pm$0.19	64.41$\pm$0.11	72.31$\pm$0.15	<u>6.74$\pm$0.03</u>	12.67

**Figure 3: ICD-9 diagnosis ontology visualization learned by different models on MIMIC-III. Best viewed in color.**

diagnosis representations on MIMIC-III using the Ward-linkage hierarchical clustering algorithm [6]. Figure 3 compares the resulting dendrograms for MedRQ and its variant that removes ontology-driven supervision (MedRQ w/o. Ontology). Additionally, we report the Silhouette score with $K = 23$, where K denotes the number of top-level concepts in the ICD-9 diagnosis ontology. A higher Silhouette score indicates better cluster compactness and separation, suggesting more semantically consistent groupings. As shown in Figure 3, without explicit guidance from the ontology, the clusters generated by MedRQ w/o. Ontology are mixed and loosely separated. This indicates that flat semantic quantization in the base RQ-VAE fails to preserve clinically meaningful hierarchical relationships. In contrast, MedRQ yields clearer and more coherent groupings that align with the multi-level ICD-9 diagnosis ontology structure, achieving a higher Silhouette score (i.e., 0.1868 vs. 0.1347). These results confirm that ontology-driven supervision enables the learned representations to capture more discriminative hierarchical structures and provide stronger interpretability.

4 Conclusion

In this paper, we present MedRQ, a novel framework that learns efficient and interpretable medical concept representations through ontology-driven residual vector quantization. The resulting quantized embeddings capture multi-level clinical hierarchies while reducing memory consumption, enabling more effective healthcare prediction. Extensive experiments demonstrate that MedRQ consistently outperforms state-of-the-art methods, highlighting its potential applicability to various healthcare prediction tasks.

Acknowledgements

This work was supported in part by the Fujian Provincial Artificial Intelligence Industry Development Technology Project under Grants (2025H0042), Fujian Provincial Natural Science Foundation of China under Grants (2025J01540), and National Natural Science Foundation of China under Grants (62302098).

References

- [1] Edward Choi, Mohammad Taha Bahadori, Jimeng Sun, Joshua Kulas, Andy Schuetz, and Walter Stewart. 2016. Retain: An interpretable predictive model for healthcare using reverse time attention mechanism. *NeurIPS* 29.
- [2] Alistair EW Johnson, Tom J Pollard, Lu Shen, Li-wei H Lehman, Mengling Feng, Mohammad Ghassemi, Benjamin Moody, Peter Szolovits, Leo Anthony Celi, and Roger G Mark. 2016. MIMIC-III, a freely accessible critical care database. *Scientific Data* 3, 1 (2016).
- [3] Alistair EW Johnson, David J Stone, Leo A Celi, and Tom J Pollard. 2018. The MIMIC Code Repository: enabling reproducibility in critical care research. *JAMIA* 25, 1 (2018).
- [4] Doyup Lee, Chiheon Kim, Saehoon Kim, Minsu Cho, and Wook-Shin Han. 2022. Autoregressive image generation using residual quantization. In *CVPR*.
- [5] Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36, 4 (2020).
- [6] Daniel Müllner. 2011. Modern hierarchical, agglomerative clustering algorithms. *arXiv* (2011).
- [7] Mohsen Nayebi Kerdabadi, Arya Hadizadeh Moghaddam, Dongjie Wang, and Zijun Yao. 2025. Multi-Ontology Integration with Dual-Axis Propagation for Medical Concept Representation. In *CIKM*.
- [8] Jesse Read, Bernhard Pfahringer, Geoff Holmes, and Eibe Frank. 2011. Classifier chains for multi-label classification. *Machine Learning* 85 (2011).
- [9] Junyuan Shang, Cao Xiao, Tengfei Ma, Hongyan Li, and Jimeng Sun. 2019. Gamenet: Graph augmented memory networks for recommending medication combination. In *AAAI*, Vol. 33.
- [10] Chaoqi Yang, Cao Xiao, Lucas Glass, and Jimeng Sun. 2021. Change Matters: Medication Change Prediction with Recurrent Residual Networks. In *IJCAI*.
- [11] Chaoqi Yang, Cao Xiao, Fenglong Ma, Lucas Glass, and Jimeng Sun. 2021. SafeDrug: Dual Molecular Graph Encoders for Recommending Effective and Safe Drug Combinations. In *IJCAI*.
- [12] Nianzu Yang, Kaipeng Zeng, Qitian Wu, and Junchi Yan. 2023. Molerec: Combinatorial drug recommendation with substructure-aware molecular representation learning. In *WWW*.
- [13] Yutao Zhang, Robert Chen, Jie Tang, Walter F Stewart, and Jimeng Sun. 2017. LEAP: learning to prescribe effective and safe treatment combinations for multimorbidity. In *KDD*.
- [14] Chuang Zhao, Hui Tang, Jiheng Zhang, and Xiaomeng Li. 2025. Unveiling Discrete Clues: Superior Healthcare Predictions for Rare Diseases. In *WWW*.
- [15] Chuang Zhao, Hongke Zhao, Xiaofang Zhou, and Xiaomeng Li. 2024. Enhancing Precision Drug Recommendations via In-depth Exploration of Motif Relationships. *TKDE* (2024).